

Lecture 26-SNMP (Simple Network Management Protocol)

1 What is SNMP?

SNMP (Simple Network Management Protocol) is an **industry-standard protocol** used to:

- Monitor network devices 
- Manage network devices 
- Control devices remotely 

SNMP allows a **central system** to check the health and performance of devices like:

- Routers
- Switches
- Firewalls
- Servers
- Printers

2 Why SNMP is Needed?

Imagine a large company network.

Example:

- 100 routers
- 200 switches
- 50 servers

Without SNMP 

A network engineer would have to:

- Log into each device manually

- Check CPU usage
- Check interface status
- Check traffic load

This is **slow and inefficient**.

With SNMP 

The monitoring system automatically:

- Collects device data
- Detects failures
- Sends alerts
- Allows remote management

 Example:

If a router interface goes **down**, SNMP immediately sends a **notification**.



SNMP History

SNMP was introduced in **1988**.

SNMP was defined using several **RFC documents**.

Important RFCs (not required to memorize):

RFC	Purpose
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RFC 1065	Structure of Management Information
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RFC 1066	Management Information Base
----------	-----------------------------

RFC 1067	SNMP Protocol
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 The name says **“Simple”**, but SNMP is very powerful.



What SNMP Can Do

SNMP can perform many management tasks:

- ✓ Monitor device status
- ✓ Monitor interface up/down
- ✓ Monitor CPU and RAM usage
- ✓ Monitor traffic load
- ✓ Detect device failures
- ✓ Send alerts (traps)
- ✓ Change configuration remotely



SNMP Architecture

SNMP uses **two main types of devices**.



Managed Devices

These are the **network devices being monitored**.

Examples:

- Routers
- Switches
- Firewalls
- Servers

Example device:

- Cisco Router
- Cisco Switch

These devices run an **SNMP Agent**.

2 Network Management Station (NMS)

This is the **central monitoring system**.

Functions:

- Sends requests to devices
- Collects information
- Receives alerts

Examples of NMS software:

- SolarWinds
- PRTG
- Nagios
- Cisco Prime

 **SolarWinds** is one of the most popular SNMP monitoring tools.

SNMP Components

Component	Description
SNMP Manager	Runs on NMS and sends requests
SNMP Agent	Runs on router/switch
SNMP Application Monitoring software	
MIB	Database of device information
OID	Unique identifier for each variable



7 Management Information Base (MIB)

MIB is a **virtual database** containing device information.

Examples of data stored:

- Interface status
- Hostname
- CPU usage
- Temperature
- Traffic counters

Important concept:

SNMP **does not store data itself**.

Instead:

SNMP **reads information from the MIB database**.



8 Object Identifier (OID)

Each piece of information in the MIB has a **unique identifier** called an **OID**.

Example OID:

1.3.6.1.2.1.1.5

OID structure looks like a **tree** .

Each number represents a **branch in the hierarchy**.

Examples:

Variable	Purpose
sysName	Router hostname
ifOperStatus	Interface operational status



SNMP Versions

● SNMPv1

- First SNMP version
 - Uses **community strings** as passwords
 - No encryption
 - Not secure ❌
-

● SNMPv2c

Improvements over SNMPv1:

- Faster performance
- Supports **GetBulk operation**
- Still uses community strings

The “c” stands for **community**.

● SNMPv3 (Best Version)

Most secure version.

Features:

- Authentication
- Encryption
- User-based security

Uses **usernames instead of community strings**.

✅ Recommended in modern networks.



10 SNMP Operations

SNMP communication uses three main operations.

Operation Sent By Purpose

GET	Manager	Read device information
SET	Manager	Change configuration
TRAP	Agent	Send alerts



1 1 SNMP Message Types

Message Type Purpose

Get	Retrieve value
GetNext	Retrieve next OID
GetBulk	Retrieve large amount of data
Set	Modify configuration
Trap	Alert message from agent
Inform	Trap with acknowledgement
Response	Reply to request



1 2 SNMP Port Numbers (VERY IMPORTANT)

Device	Port	Protocol
SNMP Agent	UDP 161	Listens for requests
SNMP Manager	UDP 162	Receives traps

1 3 SNMP Trap Example

Example situation:

Router interface **G0/1 goes down**.

What happens:

- 1 SNMP Agent detects failure
- 2 Agent creates a **trap message**
- 3 Trap is sent to **SNMP Manager**

Trap details may include:

Field	Value
Version	SNMPv2c
Community	Jeremy1
Trap Type	Interface Down
OID	Identifies the problem

The trap message is sent to **UDP port 162**.

1 4 Daily Life Example (SNMP)

Example: Internet Service Provider (ISP)

ISP network contains:

- 1000 routers
- 2000 switches

Using SNMP:

- All devices are monitored automatically
- If a fiber cable is cut 
- SNMP sends an alert immediately 

Engineers can **fix the problem quickly**.

**1****5**

SNMP LAB (Packet Tracer)



Lab Topology

PC ----- Switch ----- Router



IP Addressing

Router:

G0/0 → 192.168.1.254

PC:

IP Address → 192.168.1.1

Gateway → 192.168.1.254



Step 1 — Enable SNMP on Router

R1> enable

R1# configure terminal



Step 2 — Configure SNMP Communities

Read-Only Community

snmp-server community Cisco1 ro

Meaning:

- Allows **GET operations only**
-

Read-Write Community

snmp-server community Cisco2 rw

Meaning:

- Allows **GET and SET operations**

Step 3 — Save Configuration

do write

or

copy running-config startup-config

Step 4 — Verify SNMP

If SNMP starts successfully you may see:

%SNMP-5-WARMSTART

Meaning:

SNMP agent has started successfully.

Step 5 — Open MIB Browser on PC

In Packet Tracer:

PC → Desktop → MIB Browser

Step 6 — Configure SNMP Settings

Enter the following settings.

Field	Value
Address	192.168.1.254
Read Community	Cisco1
Write Community	Cisco2
SNMP Version	v2c

Step 7 — View Router Information

Steps:

- 1 Open **SNMP MIBs**
- 2 Open **Standard MIBs**
- 3 Select **sysName**
- 4 Click **GO**

You will see:

- OID
 - Current router hostname
 - Data type
-

Step 8 — Change Router Name (SET Operation)

Steps:

- 1 Choose **SET operation**
- 2 Data Type → **OctetString**
- 3 Value → **NewRouterName**
- 4 Click **GO**

This will change the router hostname.

Step 9 — Verify on Router CLI

Go to router CLI:

R1#

You will see the **new hostname applied**.

This confirms that **SNMP SET operation worked successfully**.

Important SNMP Commands

Command	Purpose
snmp-server community NAME ro	Read-only community
snmp-server community NAME rw	Read-write community
show snmp	SNMP status
show snmp community	View community strings
show snmp host	View trap receivers

SNMP Summary (Quick Revision)

- ✓ SNMP manages and monitors network devices
- ✓ Routers and switches are **Managed Devices**
- ✓ NMS acts as **SNMP Manager**
- ✓ Device data is stored in **MIB**
- ✓ Each variable has an **OID**
- ✓ Versions: SNMPv1, SNMPv2c, SNMPv3
- ✓ Ports used: **UDP 161 and UDP 162**

Interview One-Line Definition

“SNMP is an industry-standard protocol used to monitor and manage network devices through managers, agents, MIB databases, and OIDs.”

💡 Final Concept Memory Trick

Think of SNMP like a **hospital monitoring system** 🏥

SNMP Component Real Life Example

NMS	Doctor
Managed Device	Patient
SNMP Agent	Nurse
MIB	Medical record
OID	Patient ID

The **doctor checks all patients remotely**, just like NMS monitors network devices.



CHAPTER SUMMARY



What is SNMP?

SNMP (Simple Network Management Protocol) is an industry-standard protocol used to:

-  Monitor network devices
-  Manage network devices
-  Control devices remotely

SNMP allows a **central monitoring system** to collect information from many devices.

Devices monitored using SNMP

- Routers
- Switches
- Firewalls
- Servers
- Printers
- Wireless Controllers



Why SNMP is Needed?

Imagine a large company network.

Example:

- 100 routers
- 200 switches
- 50 servers

Without SNMP 

A network engineer must:

- Login to every device
- Check CPU manually
- Check interface status
- Check traffic load

This is **slow and inefficient**.

With SNMP

Monitoring system automatically:

- Collects device information
- Detects failures
- Sends alerts
- Allows remote management

Example

If a router interface goes down → **SNMP sends alert immediately**.

SNMP History

SNMP was introduced in **1988**.

Defined in RFC documents:

RFC	Purpose
-----	---------

RFC 1065	Structure of Management Information
----------	-------------------------------------

RFC 1066	Management Information Base
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RFC 1067	SNMP Protocol
----------	---------------

Although it is called “**Simple**”, SNMP is a **very powerful protocol**.

What SNMP Can Do

SNMP performs many monitoring tasks:

- ✓ Monitor device health
 - ✓ Monitor interface status
 - ✓ Monitor CPU usage
 - ✓ Monitor memory usage
 - ✓ Monitor traffic load
 - ✓ Detect device failures
 - ✓ Send alerts (Traps)
 - ✓ Change configuration remotely
-

SNMP Architecture

SNMP uses two main components.

1 Managed Devices

Managed devices are **network devices being monitored**.

Examples:

- Router
- Switch
- Firewall
- Server

Each managed device runs an **SNMP Agent**.

2 Network Management Station (NMS)

The **central monitoring system**.

Functions:

- Sends requests to devices
- Collects device data

- Receives alerts
- Displays monitoring dashboard

Examples of NMS software:

- SolarWinds
- PRTG
- Nagios
- Cisco Prime

🔥 **SolarWinds is one of the most popular SNMP monitoring tools.**

🔧 SNMP Components

Component	Description
SNMP Manager	Runs on NMS and sends requests
SNMP Agent	Runs on network device
SNMP Application Monitoring software	
MIB	Database of device information
OID	Unique identifier for each variable

📁 Management Information Base (MIB)

MIB is a **virtual database containing device information.**

Examples of information stored:

- Interface status
- Hostname
- CPU usage
- Temperature
- Traffic counters

Important concept:

SNMP **does not store data itself**.

Instead:

👉 SNMP reads information from the **MIB database**.

🌳 Object Identifier (OID)

Every variable in MIB has a **unique identifier** called **OID**.

Example OID:

1.3.6.1.2.1.1.5

OID structure looks like a **tree hierarchy** 🌳.

Examples:

Variable	Purpose
sysName	Router hostname
ifOperStatus	Interface operational status

🔒 SNMP Versions

● SNMPv1

First SNMP version.

Features:

- Uses **community strings** as passwords
 - No encryption
 - Not secure ❌
-

● SNMPv2c

Improved version.

Features:

- Faster performance
- Supports **GetBulk operation**
- Still uses community strings

“c” stands for **community**.

● SNMPv3 (Best Version)

Most secure version.

Features:

- Authentication
- Encryption
- User-based security

Uses **username instead of community string**.

✓ Recommended for modern networks.

✉ SNMP Operations

Operation Sent By Purpose

GET	Manager	Read information
SET	Manager	Modify configuration
TRAP	Agent	Send alerts

SNMP Message Types

Message	Purpose
---------	---------

GET	Retrieve value
-----	----------------

GETNEXT	Retrieve next OID
---------	-------------------

GETBULK	Retrieve large data
---------	---------------------

SET	Modify configuration
-----	----------------------

TRAP	Alert message
------	---------------

INFORM	Trap with acknowledgement
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RESPONSE	Reply to request
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SNMP Port Numbers (Very Important)

Device	Port	Protocol
SNMP Agent	UDP 161	Receives requests
SNMP Manager	UDP 162	Receives traps

SNMP Trap Example

Example situation:

Router interface **G0/1 goes down.**

Process:

- 1 SNMP Agent detects failure
- 2 Agent creates trap message
- 3 Trap is sent to SNMP Manager

Trap includes:

- SNMP Version
- Community string

- Trap type
- OID identifying problem

Trap is sent to **UDP port 162**.

Daily Life Example

Example: ISP Network

- 1000 routers
- 2000 switches

SNMP allows engineers to:

- Monitor devices automatically
 - Detect failures instantly
 - Fix problems quickly
-



CHAPTER CONCLUSION

SNMP is a **powerful network management protocol** used to monitor and manage devices in large networks.

Key concepts:

- ✓ SNMP enables centralized monitoring
- ✓ Managed devices run SNMP agents
- ✓ NMS acts as SNMP manager
- ✓ MIB stores device data
- ✓ OIDs uniquely identify variables
- ✓ SNMP uses UDP ports **161 and 162**
- ✓ SNMPv3 is the most secure version

SNMP is widely used in:

- Enterprise networks
- Internet service providers

- Data centers
- Cloud infrastructure

Understanding SNMP is **essential for network engineers and NOC engineers**.

Q & A

1 What is SNMP?

SNMP (Simple Network Management Protocol) is a network management protocol used to **monitor, manage, and control network devices** from a central management system.

It allows administrators to **collect performance data, detect failures, and configure devices remotely**.

2 Why is SNMP used?

SNMP is used to **simplify network monitoring**.

Without SNMP, engineers must log into each device manually.

With SNMP:

- Device status is monitored automatically
 - Alerts are generated when problems occur
 - Performance metrics are collected centrally
-

3 What are managed devices?

Managed devices are **network devices being monitored using SNMP**.

Examples:

- Routers
- Switches
- Firewalls

- Servers

Each device runs an **SNMP agent**.

What is an SNMP Agent?

An SNMP Agent is software running on a managed device.

Functions:

- Collect device information
 - Store information in MIB
 - Respond to SNMP manager requests
 - Send trap alerts
-

What is NMS?

NMS (Network Management System) is a **central monitoring platform** that communicates with SNMP agents.

Examples:

- SolarWinds
 - PRTG
 - Nagios
-

What is MIB?

MIB (Management Information Base) is a **database containing device management information**.

Examples of information:

- CPU usage
- Memory usage
- Interface status
- Device hostname

7 What is OID?

OID (Object Identifier) is a **unique identifier used to access specific information in the MIB database.**

Example:

1.3.6.1.2.1.1.5

This may represent the **device hostname.**

8 What are SNMP versions?

Three main versions:

Version	Description
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SNMPv1	First version, no security
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SNMPv2c	Improved performance
---------	----------------------

SNMPv3	Secure version with encryption
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9 Which SNMP version is most secure?

SNMPv3 is the most secure version.

Features:

- Authentication
 - Encryption
 - User-based access control
-

10 What are SNMP operations?

Three main operations:

GET – retrieve device information

SET – change configuration

TRAP – alert notification

1 1 What is an SNMP Trap?

An SNMP Trap is a **notification sent by the SNMP agent to the SNMP manager when a specific event occurs.**

Example events:

- Interface down
 - CPU overload
 - Device reboot
-

1 2 What ports does SNMP use?

SNMP uses UDP protocol.

Port	Purpose
------	---------

UDP 161	SNMP requests
---------	---------------

UDP 162	SNMP traps
---------	------------

1 3 Difference between SNMP Trap and Inform?

Trap:

- One-way notification
- No acknowledgement

Inform:

- Trap with acknowledgement

- Manager confirms receipt
-

1 4 How do you configure SNMP on Cisco router?

Example configuration:

```
snmp-server community Cisco1 ro
snmp-server community Cisco2 rw
```

1 5 How do you verify SNMP configuration?

Commands:

```
show snmp
show snmp community
show snmp host
```

1 6 Practical Interview Question

Scenario:

Router interface goes down.

What happens?

Answer:

- SNMP agent detects event
 - Agent generates trap
 - Trap sent to NMS
 - Engineer receives alert
-